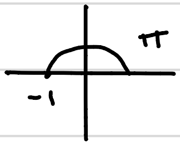
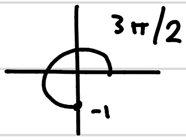
MTH2003M   Practical 1

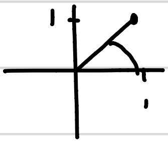
	0	$\pi/6$	$\pi/4$	$\pi/3$	$\pi/2$
sin	0	$1/2$	$1/\sqrt{2}$	$\sqrt{3}/2$	1
cos	1	$\sqrt{3}/2$	$1/\sqrt{2}$	$1/2$	0
tan	0	$1/\sqrt{3}$	1	$\sqrt{3}$	$*$

a) 

$$-1 = e^{i\pi}$$

b) 

$$-i = e^{3i\pi/2}$$

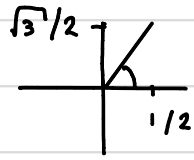
c) 

$$|z| = \sqrt{1^2 + 1^2} = \sqrt{2}$$

$$\theta = \tan^{-1}(1) = \pi/4 \text{ or } 5\pi/4$$

Angle lies in first quadrant, so $\theta = \pi/4$, and

$$z = \sqrt{2}\, e^{i\pi/4}$$

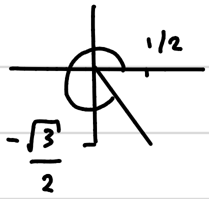
d) 

$$|z| = \sqrt{\left(\frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2} = \sqrt{\frac{1}{4} + \frac{3}{4}} = 1$$

$$\theta = \tan^{-1}(\sqrt{3}) = \pi/3 \text{ or } 4\pi/3$$

Angle lies in the first quadrant, so
$\theta = \pi/3$, and

$$z = e^{i\pi/3}$$

e) 

$|z| = 1$, as above

and $\theta = \tan^{-1}(-\sqrt{3})$
$= 2\pi/3$ or $5\pi/3$

Angle lies in 4th quadrant, so $\theta = 5\pi/3$
and

$$z = e^{5\pi i/3}$$

2 a) $e^{2 + i\pi/2} = e^{2} e^{i\pi/2} = e^{2} i$

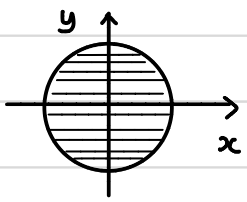
b) $\dfrac{1}{i} = \dfrac{i}{i.(-i)} = \dfrac{-i}{1} = -i$

c) $\dfrac{1}{1+i} = \dfrac{1}{1+i} \dfrac{(1-i)}{(1-i)} = \dfrac{1-i}{1^2+1^2} = \dfrac{1}{2} - \dfrac{i}{2}$

d) $(1+i)^3 = 1 + 3i + 3i^2 + i^3$
$= 1 + 3i - 3 - i$
$= -2 + 2i$

e) $|3 + 4i| = (3^2 + 4^2)^{1/2} = (9 + 16)^{1/2} = 25^{1/2} = 5$

3 a) $|z| \leqslant 1 \Rightarrow |z|^2 \leqslant 1$
$\Rightarrow x^2 + y^2 \leqslant 1$



Disk of radius 1 centred at the origin

Note: we can square both sides of the inequality, since $f(x) = x^2$ is an increasing function of x for $x > 0$, and, if $x_2 > x_1$, then $x_2^2 > x_1^2$.

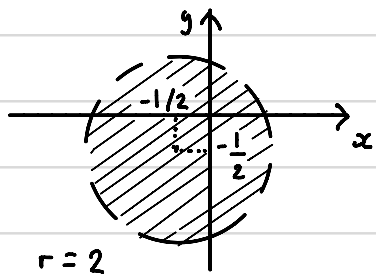
b) $|2z + 1 + i| < 4$, so $\left|z + \frac{1}{2} + \frac{i}{2}\right| < 2$

Then, $\left|x + iy + \frac{1}{2} + \frac{i}{2}\right| < 2$

and $|(x + 1/2) + i(y + 1/2)| < 2$

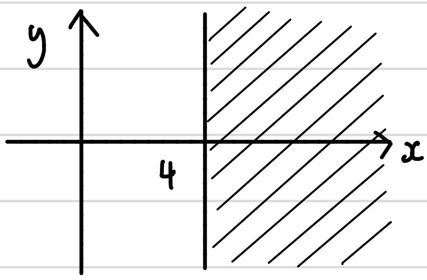
so that $(x + 1/2)^2 + (y + 1/2)^2 < 4$

Disk of radius 2 centred at $(-1/2, -1/2)$.



(dashed line used to show that boundary is not included)

c)



$Re(z) = x \geq 4$.

d) $|z| \leq |z + 1|$, so $|z|^2 \leq |z + 1|^2$

Then, $|x + iy|^2 \leq |x + iy + 1|^2$

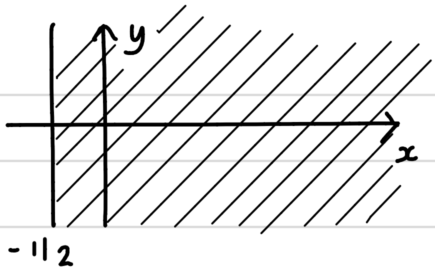
$$|x + iy|^2 \leq |(x+1) + iy|^2$$

$$x^2 + y^2 \leq (x + 1)^2 + y^2$$

$$x^2 + y^2 \leq x^2 + 2x + 1 + y^2$$

$$0 \leq 2x+1$$

$$-1/2 \leq x$$

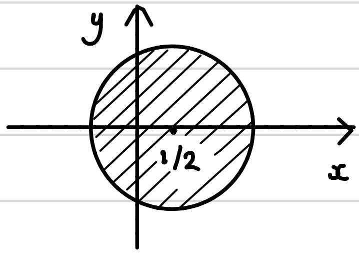


e) $0 < |2z - 1| \leq 2$

$0 < |z - 1/2| \leq 1$

$0 < |x + iy - 1/2| \leq 1$

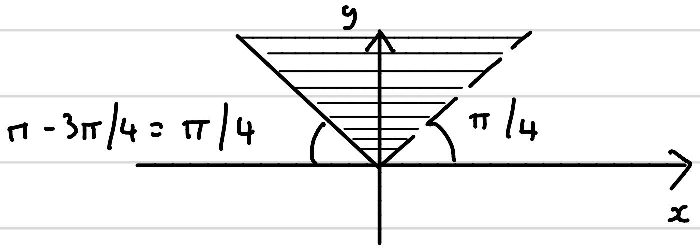
$0 < |(x - 1/2) + iy| \leq 1$



$$0 < (x - 1/2)^2 + y^2 \leq 1$$

Punctured disk with radius 1 and centre at $x = 1/2$.

f)



4. Let $z = a + ib$ and $w = c + id$

$$\overline{(z + w)} = \overline{(a + ib + c + id)} = \overline{[(a+c) + i(b+d)]}$$

$$= (a+c) - i(b+d) = (a - ib) + (c - id)$$

$$= \bar{z} + \bar{w}.$$

5.

$$\overline{zw} = \overline{(a+ib)(c+id)} = \overline{[(ac-bd) + i(ad+bc)]}$$

$$= (ac - bd) - i(ad + bc)$$

Now, $\bar{z}\bar{\omega} = (a - ib)(c - id) = (ac - bd) - i(ad + bc)$,

which is equal to the expression for $\overline{z\omega}$ found above.

5. $z = x + iy$ and $\bar{z} = x - iy$

Then, $z + \bar{z} = 2x$, and $x = \mathrm{Re}(z) = \frac{1}{2}(z + \bar{z})$

Also, $z - \bar{z} = 2iy$, so $y = \mathrm{Im}(z) = \frac{1}{2i}(z - \bar{z})$

Using the above formulas in the equation

$x + 3y = 2$, we find

$$\frac{1}{2}(z + \bar{z}) + \frac{3}{2i}(z - \bar{z}) = 2$$

and $\left(\frac{1}{2} + \frac{3}{2i}\right) z + \left(\frac{1}{2} - \frac{3}{2i}\right)\bar{z} = 2$

or $(3 + i) z + (-3 + i)\bar{z} = 4i$

6. We are solving $z^3 = i$.

Writing $z = re^{i\theta}$ and $i = e^{i\pi/2 + 2k\pi i}$, $k \in \mathbb{Z}$

we find $r^3 e^{3\theta i} = 1 . e^{i\pi/2 + 2k\pi i}$

and $re^{\theta i} = 1^{1/3} e^{i\pi/6 + 2k\pi i/3}$

so that $r = 1^{1/3} = 1$ and $\theta = \pi/6$, $5\pi/6$
or $3\pi/2$

The three roots are then

$$z_0 = 1 . e^{\pi i/6} = \cos(\pi/6) + i \sin(\pi/6)$$
$$= \frac{\sqrt{3}}{2} + \frac{i}{2}$$

$$z_1 = 1. e^{5\pi i/6} = \cos\left(\frac{5\pi}{6}\right) + i\sin\left(\frac{5\pi}{6}\right) = -\frac{\sqrt{3}}{2} + \frac{i}{2}$$

$$z_2 = 1 . e^{3\pi i/2} = \cos\frac{3\pi}{2} + i \sin\frac{3\pi}{2} = -i$$

7 a) $z^3 = 4$, or $r^3 e^{3i\theta} = 4 e^{2k\pi i}$, $k \in \mathbb{Z}$.

Then, $re^{i\theta} = 4^{1/3} e^{2k\pi i/3}$, so that $r = 4^{1/3}$ and $\theta = 0, 2\pi/3$ and $4\pi/3$.

The three roots are

$$z_0 = 4^{1/3}, \quad z_1 = 4^{1/3} e^{2\pi i/3}, \quad z_2 = 4^{1/3} e^{4\pi i/3}$$

b) $z^4 = -1$, or $r^4 e^{4i\theta} = 1 e^{i\pi + 2k\pi i}$, $k \in \mathbb{Z}$.

Then, $re^{i\theta} = 1^{1/4} e^{i\pi/4 + k\pi i/2}$, so that

$r = 1$ and $\theta = \pi/4, 3\pi/4, 5\pi/4, 7\pi/4$

and $z_0 = e^{i\pi/4}$, $z_1 = e^{3\pi i/4}$, $z_2 = e^{5\pi i/4}$,

$z_3 = e^{7\pi i/4}$.

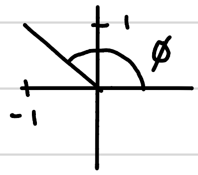
c) $z^4 + 2z^2 + 2 = 0$

Quadratic in z^2: $z^2 = \dfrac{-2 \pm \sqrt{2^2 - 4 \times 2}}{2}$

$= -1 \pm i$

- First case: $|-1+i| = (1^2 + 1^2)^{1/2} = \sqrt{2}$

$\phi = \tan^{-1}(1/(-1)) = 3\pi/4$ or $7\pi/4$.



Second quadrant, so $\phi = 3\pi/4$.

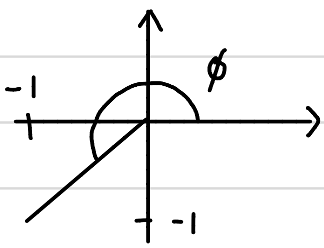
Then, $z^2 = -1 + i$ becomes

$$r^2 e^{2i\theta} = 2^{1/2} e^{3\pi i/4 + 2k\pi i}, \quad k \in \mathbb{Z}$$

Next, $r^2 = 2^{1/2}$, so $r = 2^{1/4}$, and $2\theta = 3\pi/4 + 2k\pi$.
Then, $\theta = 3\pi/8 + k\pi$, and $\theta = 3\pi/8$ or $11\pi/8$.

- Second case: $|-1-i| = (1+1)^{1/2} = 2^{1/2}$

$\phi = \tan^{-1}(-1/(-1)) = \pi/4$ or $5\pi/4$



Third quadrant, so $\phi = 5\pi/4$.

Then, $z^2 = -1 - i$ becomes

$$r^2 e^{2i\theta} = 2^{1/2} e^{5\pi i/4 + 2k\pi i}, \quad k \in \mathbb{Z}.$$

Next, $r^2 = 2^{1/2}$, so $r = 2^{1/4}$, and $\theta = 5\pi/8$ or $13\pi/8$

The four roots are then

$$2^{1/4} e^{3\pi i/8}, \quad 2^{1/4} e^{11\pi i/8}, \quad 2^{1/4} e^{5\pi i/8}, \quad 2^{1/4} e^{13\pi i/8}$$

8. $$(az+b)^3 = c$$
$$= c e^{2\pi k i}, \quad k \in \mathbb{Z}$$

Then, $az + b = c^{1/3} e^{2\pi k i/3}$

and $$z = \frac{c^{1/3} e^{2\pi i k/3} - b}{a}$$

The three roots are $\dfrac{c^{1/3} - b}{a}$, $\dfrac{c^{1/3} e^{2\pi i/3} - b}{a}$

and $\dfrac{c^{1/3} e^{4\pi i/3} - b}{a}$